

FIG. 11: Our Replica Electromagnet

To confirm some of the suggestions made about Henry's experiments, we constructed a replica of Henry's Albany magnet, shown in Figure 11 above. Each coil was wound in the same way as experiments 8-16. There are a couple of main differences between our magnet and Henry's: 1) Our magnet was forged from a mild steel, which has a higher carbon content than wrought iron, giving it some residual magnetism after the external field is removed. Mild steel also takes much longer to reach saturation, and has a smaller initial permeability; 2) We used 9, 60 ft strands of high purity enameled copper wire, which has a much thinner insulation than Henry's silk, allowing for a greater number of turns, and has a lower resistivity than the copper wire used by Henry. These two differences, especially using a different magnetic material, can be shown to have a large effect on our results. Nevertheless, all the data collected below should be consistent enough to establish certain comparisons with Henry's results.

When measuring the B-field at different areas on the poles, it was observed that the B-Field increased when the probe was brought towards the inner edge of the pole. Furthermore, the B-field was at a maximum at the inner corners of each pole (about double that of the center). Even though the reading was higher at the corners, it was difficult to get consistent measurements there, so these results were discarded.

To test the idea of flux leakage through a magnetic circuit, each coil was attached to a constant current of 5 Amps and the B-Field was measured at the center of each pole. The results are shown below in Table I.

Coil	Right side B-Field (Gauss)	Left Side B-Field (Gauss)	Total B-Field (Gauss)
1	140	29	169
2	117	49	166
3	98	53	151
4	87	68	155
5	78	78	156
6	58	97	155
7	48	107	155
8	39	112	151
9	29	136	165

TABLE I: B- Field measurements taken for each coil supplied with a constant current